

2024



Jungheinrich Optimizes Intralogistics Using Shop Floor Connectivity by AWS

Learn how intralogistics company Jungheinrich is facilitating advanced analytics on its equipment using AWS.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#) | [Architecture Diagram](#)

Nearly 25% increase
in sampling rate

0.01% instead of 3% proportion
of data gaps

Overview

Since 2019, global logistics have been in the spotlight by the COVID-19 pandemic and geopolitical expectations for shorter and more reliable supply chains is fundamental for success.

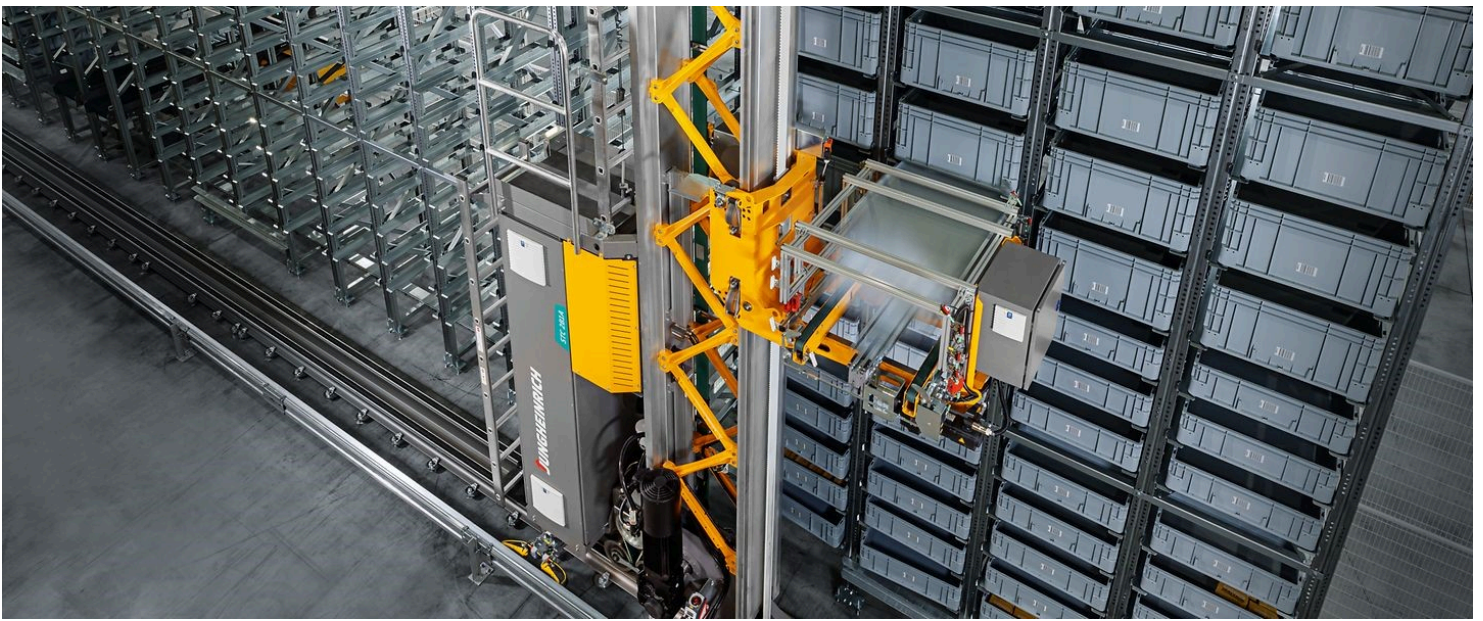


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Opportunity | Using AWS to Simplify Machine Connectivity for Jungheinrich

[Jungheinrich](#), a global leader in intralogistics solutions, uses digital technology to provide its customers with even more efficient and reliable solutions. The company wanted to gain near real-time visibility, equipment insights, and predictive analytics for one highly sophisticated product line: stacker cranes. But connecting machines, especially at customers' sites, is a complex task. This process often requires bespoke configuration and specialized personnel.

Jungheinrich deployed an industrial edge offering to overcome these challenges: [Shop Floor Connectivity \(SFC\)](#), an open-source software framework for machine connectivity. This software can connect machines, collect and analyze sensor data locally at high frequency, and ingest aggregated data with context to the cloud. The collected data is processed in near real time to determine the condition of the equipment, predict potential failures, and optimize operations.

Solution | Optimizing Industrial Edge Configuration on AWS

Using the open-source SFC in conjunction with Amazon Web Services (AWS) offerings like [AWS IoT](#), which helps organizations deploy and configure software to the industrial edge, Jungheinrich simplified the deployment of new machines and significantly improved data quality. The sampling rate has increased by 25 percent, and the proportion of data gaps has been reduced from 3 percent to 0.01 percent.



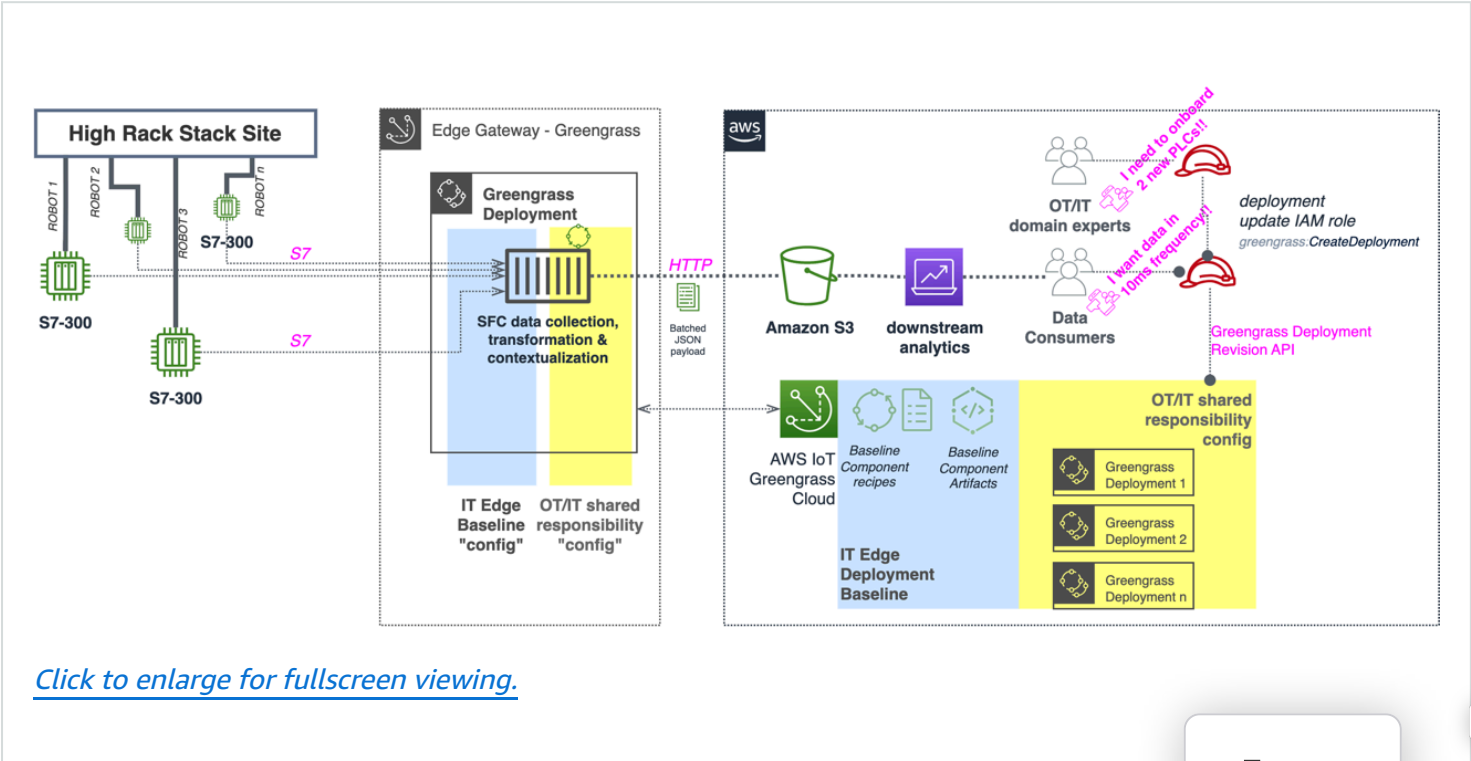
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Jungheinrich deployed SFC to an industrial PC powered by AWS IoT Greengrass in a matter of days. This offering supports various Industrial Ethernet-based protocols like Siemens S7, Beckhoff ADS, and Rockwell Automation PCCC. It also integrates with AWS services like [AWS IoT Core](#), which connects billions of IoT devices and routes trillions of messages.

Jungheinrich configured the solution to connect and ingest data from automation equipment and sensors—no coding required. Equipment experts from the shop floor can quickly and efficiently onboard new equipment. Onboarding new equipment is now just one API call away, accelerating several potential new use cases, such as predictive analytics. Additionally, data analysts now have flexible access to raw machine data to perform advanced analytics and develop purpose-built algorithms.



Architecture Diagram



AWS IoT Greengrass and SFC laid the foundation for Jungheinrich to explore proofs of concept and develop digital products for automated warehouse systems. The company is now enhancing its condition-based maintenance solution on AWS, which will provide technicians with maintenance recommendations based on data analysis. Preventive maintenance will make way for predictive maintenance, reducing downtimes and enhancing asset availability and productivity at customer sites. Thanks to these capabilities, Jungheinrich’s service and support will become more efficient and effective.

About Jungheinrich

Jungheinrich is one of the world’s leading companies in the material handling equipment industry, with a significant global presence. It has over 18,000 employees worldwide and operations in more than 40 countries.

AWS Services Used

AWS IoT Greengrass

AWS IoT Greengrass is an open-source edge runtime and cloud service for building, deploying, and managing device software.

[Learn more »](#)

Amazon IoT Core

AWS IoT Core lets you connect billions of IoT devices and route trillions of messages to AWS services without managing infrastructure.

[Learn more »](#)

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